

Anchored Relational Residual Adaptation for CLIP

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Abstract

CLIP-style dual encoders support open-vocabulary retrieval and zero-shot recognition, but they often score captions by object co-occurrence while underweighting relation, role, and binding edits. We study whether relation sensitivity can be added without replacing the pretrained embedding path. Relational Residual Bottlenecking (RRB) freezes OpenCLIP ViT-B/16 and learns a bounded, parser-free residual correction, $z = \text{normalize}(z_0 + \alpha\delta_{\text{rel}})$, from multi-channel token bottlenecks on each branch. On staged ARO/SugarCrepe relation suites, the selected no-targeted strong-anchor checkpoint improves relation accuracy from the plain-adaptor baseline’s 0.6956 to 0.7734 while retaining COCO image-to-text R@1 of 0.7340 and ImageNet top-1 of 0.6342. Hard token bottlenecks and no-channel residual controls achieve relation gains only with much larger retrieval and zero-shot degradation. The experiments also give a useful negative result: structured relational counterfactuals matter, while the earlier targeted slot loss is not necessary. The claim is deliberately bounded. COCO retention remains seed-fragile, Winoground transfer is flat-to-negative, Flickr30k shows an image-to-text retrieval tax, and responsibility probes do not support semantic slot specialization.

CCS Concepts

• **Computing methodologies** → **Computer vision**; • **Information systems** → *Multimedia and multimodal retrieval*.

Keywords

vision-language models, CLIP, compositionality, residual adaptation, retrieval

1 Introduction

Contrastive image-text encoders such as CLIP are now standard infrastructure for open-vocabulary retrieval, zero-shot classification, and larger multimodal systems [5, 16]. Their shared embedding space makes retrieval efficient, but it can underweight relational edits: captions that preserve object vocabulary while changing a role, spatial relation, attribute, or word order may remain too close in the embedding space [3, 20, 25]. For information retrieval, this failure mode

matters because a visually plausible but relationally wrong caption can be ranked above the correct one.

This report asks a narrow research question: can a frozen CLIP-style dual encoder be made more sensitive to relational counterfactuals without replacing the pretrained embedding path? We use “guardrail” in a technical sense only: retention of retrieval and zero-shot behavior on held-out COCO and ImageNet checks. The central novelty is not relation supervision for CLIP in general. It is the combination of a bounded residual bottleneck interface and a guardrail-centered empirical analysis of how relation gains trade off against preservation of the original OpenCLIP geometry.

The design pressure is simple. If a method obtains higher relation accuracy by replacing the frozen embedding with a new representation, then the relation score may only show that the method learned the staged benchmark. It does not show that the adapted model remains useful as a CLIP retrieval encoder. RRB instead keeps the frozen OpenCLIP representation available to every downstream dot product and learns a small residual correction. A relation gain is therefore necessary but not sufficient; it must be read together with retrieval, zero-shot, residual-size, and external-validity checks.

The supported claim is deliberately bounded: anchored residual adaptation can improve staged relational sensitivity in a frozen CLIP-style dual encoder while bounding, but not eliminating, retrieval and zero-shot degradation. The selected no-targeted, stronger-anchor checkpoint improves relation accuracy from the plain adaptor baseline’s 0.6956 to 0.7734, while retaining COCO image-to-text R@1 of 0.7340 and ImageNet top-1 of 0.6342. This is not lossless preservation. It is a better tradeoff than the hard-token and no-channel controls, which reach relation gains only with much larger retrieval and zero-shot damage.

The project also reports negative findings because they are part of the new knowledge learned from the study. Structured relational counterfactuals are the useful training signal; generic hard negatives and the earlier targeted slot loss are not sufficient explanations. Hard bottlenecks and no-channel residual controls show that relation accuracy is easy to improve by damaging the embedding geometry. COCO retention remains seed-fragile, Winoground transfer is flat-to-negative, Flickr30k shows an image-to-text retrieval tax, and

responsibility probes do not support semantic slot specialization. RRB should therefore be read as a residual-interface tradeoff analysis, not as a claim that it solves compositional reasoning or dominates standard PEFT methods.

We make four contributions.

- (1) We define Relational Residual Bottlenecking, a parser-free bounded residual module for frozen OpenCLIP image and text encoders.
- (2) We show that the selected RRB checkpoint improves staged relation accuracy while avoiding the catastrophic guardrail collapse of hard replacement bottlenecks and no-channel residual controls.
- (3) We isolate the central training signal: structured relational counterfactuals under residual anchoring, rather than generic hard negatives or the earlier targeted relational loss.
- (4) We document the limitations that bound the claim: seed-fragile COCO retention, negative or mixed external checks, and diffuse channel responsibility.

The rest of the report follows this claim boundary. Section 2 positions RRB relative to compositional CLIP evaluation, structure-aware retrieval, and preservation-oriented adaptation. Section 3 defines the residual interface and objective. Section 4 reports the empirical tradeoffs. Section 5 analyzes challenges, limitations, and lessons learned before the conclusion and reproducibility notes.

2 Related Work

Compositional failures in CLIP. CLIP and OpenCLIP remain common frozen backbones for retrieval and zero-shot transfer [5, 16], but several benchmarks show that strong image-text encoders can behave like bags of words when attributes, relations, or word order change [3, 20, 25]. We use ARO- and SugarCrepes-style counterfactual suites because they isolate exactly this failure mode. We also include Winoground and Flickr30k follow-ups because success on staged caption edits does not by itself imply broad binding generalization or lossless retrieval preservation.

Structured image-text matching. Graph and structure-aware retrieval methods use scene graphs, relationship-enhanced semantic graphs, multi-level semantic consistency, triple-level relation reasoning, and cross-modal graph reasoning to make image-text relations explicit [8, 9, 12, 19, 22]. Spatial and knowledge-oriented variants similarly add structural cues through multimodal knowledge graphs, 3D scene graphs, or spatial supervision [1, 2, 7, 17]. Taken together, this line of work exposes a recurring gap: explicit structure helps relation modeling, but it often changes the input representation, adds graph or knowledge construction, or moves matching away from the plain CLIP dual-encoder interface. RRB keeps the deployed interface closer to CLIP: images and texts are encoded independently, no detector or parser is used at inference, and retrieval remains a dot product between one image vector and one text vector.

Relation and reasoning benchmarks. Broader multimodal reasoning work probes relation composition through scene-graph prompting, multi-image relational benchmarks, visual causal reasoning, and causal-order diagnostics [6, 13, 23, 26]. These papers motivate relation-sensitive evaluation, but they often target large multimodal model reasoning or multi-image settings rather than frozen dual-encoder retrieval. We therefore use them as external context for why staged relation gains are not enough, and we rely on Winoground and Flickr30k checks to bound our own claim.

Relation-aware CLIP adaptation. Relation-aware CLIP work, including the local RACA-CLIP manuscript available in this project, targets compositional relation sensitivity directly [21]. We treat that manuscript as conceptual prior art with conservative metadata because the project copy does not provide full public bibliographic details. RRB differs in emphasis: relation supervision is routed through a bounded residual correction to the frozen embedding, and the evaluation centers on the relation-vs-preservation tradeoff rather than relation accuracy alone.

Preservation-oriented adaptation. Residual adapters, low-rank adaptation, null-space projection, gradient projection, residual-gradient adaptation, and principal-subspace constraints are natural ways to adapt pretrained models while limiting drift [4, 10, 11, 14, 15, 24]. We do not claim that residual adaptation or orthogonal projection is novel. The supported contribution is narrower: structured relational counterfactual training through a bounded multi-channel residual interface, plus diagnostics showing when that interface helps and when it fails. Standard PEFT baselines such as LoRA, text-only tuning, vision-only tuning, and prompt tuning were not run, so the paper does not claim that RRB dominates them.

3 Relational Residual Bottlenecking

3.1 Setup

Let f_v and f_t be frozen OpenCLIP image and text encoders. For an image I or caption T , the frozen model produces normalized embeddings $z_0^v = f_v(I)$ and $z_0^t = f_t(T)$. The goal is to adjust the embedding used for contrastive comparison without discarding z_0 , since the pretrained path carries the broad retrieval and zero-shot behavior that we want to retain.

The method is designed for the dual-encoder setting. At inference time, images and captions must still be embeddable independently, and the final similarity remains a dot product between normalized vectors. No detector, parser, cross-attention reranker, scene graph, or candidate-pair module is added. This choice rules out some stronger reasoning machinery, but it keeps the evaluation focused on whether a pretrained embedding can be corrected rather than replaced.

RRB attaches a small trainable residual module to each branch:

$$z^b = \text{normalize} \left(z_0^b + \alpha \delta_{\text{rel}}^b \right), \quad b \in \{v, t\}. \quad (1)$$

Here δ_{rel}^b is computed from CLIP token features through a parser-free multi-channel bottleneck and α is a fixed residual gate. The selected checkpoint uses $\alpha = 0.05$. The residual therefore has to act as a correction to the original representation rather than a replacement representation.

The fixed gate is deliberately simple. A learned gate can increase flexibility, but the early R022 diagnostic run, omitted from the main tables for brevity, showed that more flexible residual control can still miss the COCO gate. A fixed gate is preferred because it maintains the architectural prior of a dominant pretrained path throughout training, preventing the residual branch from taking over the embedding space through an unconstrained mixing coefficient. The selected method fixes the residual scale and lets the objective decide when the residual direction is useful. This makes the resulting claims easier to audit: if guardrails fail, the failure cannot be attributed to an unbounded learned mixing coefficient silently replacing the frozen path.

3.2 Multi-Channel Residual Interface

For each branch, the module in Figure 1 receives the frozen token sequence and induces K latent channel states through learned queries and one self-attention block. The channel states are pooled and projected to the CLIP embedding dimension, producing δ_{rel}^b . We avoid the term “semantic slots” for the main method because later responsibility probes do not show semantic specialization. The bottleneck is used as an architectural constraint on the residual path, not as an interpretability claim.

The no-channel controls are important because they isolate this interface. R033 and R040 remove the multi-channel bottleneck and use a simpler residual adapter. Both remain relation-strong but damage retrieval and zero-shot behavior much more than RRB, suggesting that the bottlenecked residual interface is preservation-relevant even though its channels are not semantically interpretable.

The interface can be understood as a capacity and routing prior. The residual is not produced by one unconstrained multilayer projection from all token features to the final embedding. Instead, the token sequence must be summarized through a small number of channel states before it can affect the CLIP vector. This is not a guarantee of safety, and the seed results show that it is not enough by itself. It does, however, create a narrower path for the relation objective to act through. The contrast between R028 and the no-channel controls is consistent with this interpretation: similar or stronger relation accuracy can be achieved without the channel interface, but the resulting embedding drift is far more harmful to COCO and ImageNet.

This caution is consistent with recent analyses of ViT token aggregation, which show that global tokens can rely on background-dominated shortcut patches under coarse supervision [18]. RRB does not attempt to fix dense-feature artifacts. The relevance is narrower: token summaries are not automatically benign, so the paper evaluates the final residual

direction through retrieval and zero-shot guardrails instead of assuming that a small token-derived correction is safe.

The paper therefore separates three ideas that are easy to conflate. First, multiple channels are an architectural bottleneck. Second, the channels might become specialized in some future model. Third, the current probes do not show that specialization. Only the first statement is part of the method claim. The second is a hypothesis, and the third is a negative result reported in Section 4.

3.3 Training Objective

Training uses structured relation counterfactuals and anchor examples:

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{anchor}}\mathcal{L}_{\text{anchor}} + \lambda_{\Delta}\mathcal{L}_{\Delta}. \quad (2)$$

$\mathcal{L}_{\text{rel}}$ is a margin ranking loss over structured relation edits. Given a positive pair (I, T^+) and a counterfactual caption T^- that changes a relation, role, order, object, or attribute while preserving much of the surface vocabulary, the loss encourages $\text{sim}(z_I^v, z_{T^+}^t)$ to exceed $\text{sim}(z_I^v, z_{T^-}^t)$ by a margin, where sim is the dot product between normalized embeddings. The same principle applies to branch-specific counterfactuals where available.

$\mathcal{L}_{\text{anchor}}$ is feature mimicry on anchor pairs:

$$\mathcal{L}_{\text{anchor}} = \|z^v - z_0^v\|_2^2 + \|z^t - z_0^t\|_2^2. \quad (3)$$

It keeps the adapted embeddings close to the frozen OpenCLIP anchor on examples that should not require relational correction. $\mathcal{L}_{\Delta}$ penalizes residual magnitude,

$$\mathcal{L}_{\Delta} = \|\delta_{\text{rel}}^v\|_2^2 + \|\delta_{\text{rel}}^t\|_2^2, \quad (4)$$

which works with the fixed gate in Eq. 1 to bound feature drift.

An earlier targeted relational loss attempted to localize edited spans to particular bottleneck channels. Unlike $\mathcal{L}_{\text{rel}}$, which directly ranks the positive caption above a structured counterfactual in the final embedding space, this auxiliary term compared the channel states for the positive and edited captions and penalized diffuse or mislocalized channel changes. The final method removes that term. The ablations show that structured counterfactual supervision and residual anchoring are sufficient for the observed relation gain; the targeted loss is not necessary and should not be used as a mechanism claim.

3.4 Structured Counterfactual Signal

The relation loss depends on counterfactual captions whose surface form is close to the positive caption but whose relation, role, object, attribute, or order is changed. This structure is what distinguishes the method from generic hard-negative mining. Generic negatives can be visually or semantically difficult, but they need not isolate the binding error that CLIP tends to miss. In R024, generic hard negatives preserve broad OpenCLIP behavior almost perfectly, yet relation accuracy falls below the plain adapter baseline. That run is important because it rules out a simpler explanation:

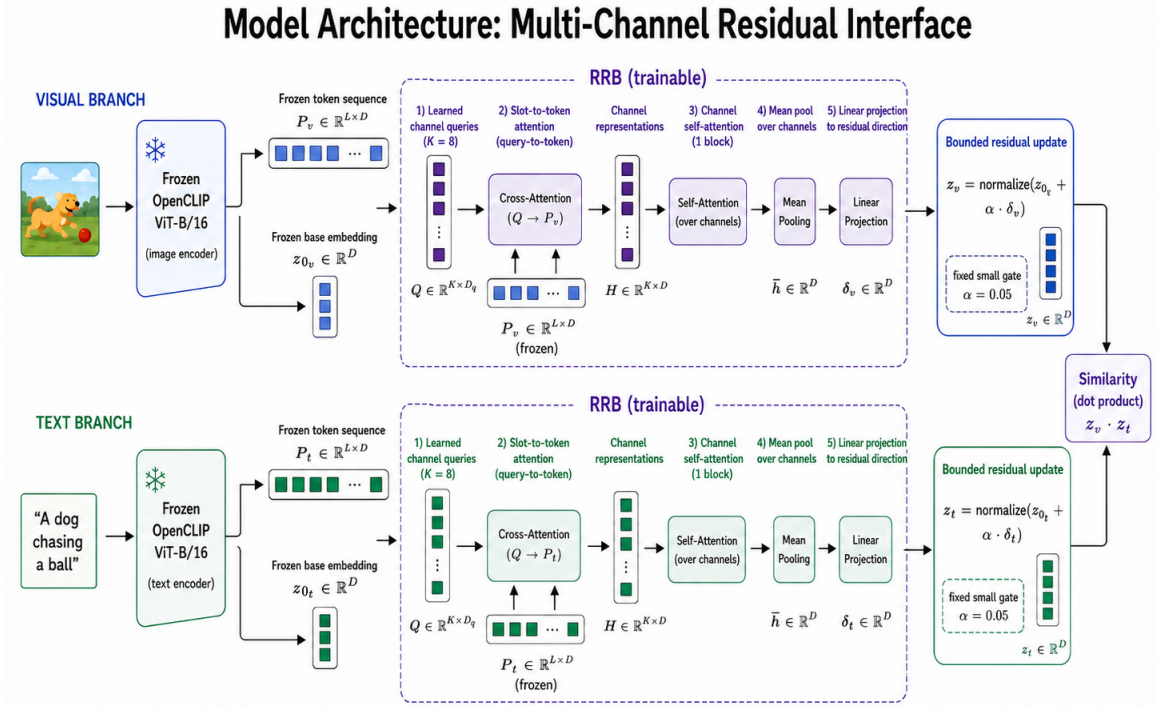


Figure 1: RRB module used on both image and text branches. Frozen OpenCLIP token features are routed through $K = 8$ learned channel queries, channel self-attention, pooling, and a linear projection to produce a residual direction. The fixed gate $\alpha = 0.05$ keeps the residual correction bounded before the final normalized embedding is used for dot-product retrieval.

the selected model is not merely benefiting from more difficult negatives. It benefits from negatives that target the intended failure mode.

The training signal is also deliberately branch-compatible. Both image and text branches keep their frozen encoders, and both branches receive residual modules. The margin objective is evaluated through the final normalized embeddings, so the residual is rewarded only when it improves the objective aligned with the deployed interface, although it also means that benchmark-specific counterfactual patterns can be learned without broader binding generalization. The Winoground and Flickr30k checks are included to expose that risk.

3.5 Anchor Examples and Residual Regularization

The anchor term uses examples that should not require a relational correction and asks the adapted embedding to mimic the frozen OpenCLIP embedding. This term is not a generic

preference for small weights; it is a feature-space constraint at the point where downstream retrieval operates. The residual penalty is complementary. $\mathcal{L}_{\text{anchor}}$ constrains the final normalized output on anchor examples, while $\mathcal{L}_{\Delta}$ constrains the branch residual itself. In the selected R028 configuration, $\lambda_{\text{anchor}} = 10$, $\lambda_{\Delta} = 0.01$, and the anchor ratio is 0.75.

The ablations show why both the anchor objective and the residual direction matter. The no-targeted weak-anchor configuration (R023) has low residual magnitude and high cosine to the frozen embedding, but it still misses the COCO gate. The selected stronger-anchor no-targeted configuration (R028) keeps a similar small-residual regime and passes the pilot gate. The matched stronger-anchor targeted control (R029) reintroduces the targeted loss and has a larger residual than R028 with no relation advantage. Anchor-20 runs reduce residual magnitude further but do not eliminate COCO seed fragility. The method should therefore be read as a bounded residual recipe, not as a theorem that smaller residuals are always safer.

3.6 Model Selection Rule

During experimentation, candidate checkpoints were evaluated with a pilot gate: relation accuracy had to exceed the plain adapter, COCO image-to-text R@1 had to be at least 0.70, and ImageNet top-1 had to be at least 0.55. This gate is not a universal deployment criterion. It is a project-level guard against selecting a model that wins the staged relation suite by destroying broad alignment. R028 is selected because it satisfies this gate while removing the unsupported targeted loss from the method. The seed studies then test how stable that selection is, rather than treating the selected checkpoint as sufficient evidence by itself.

For any practical use, this gate should be applied as a validation protocol rather than as a one-shot model choice. Multiple seeds should be trained under the same configuration; checkpoints that fail the relation, COCO, or ImageNet gates should be discarded; and the reported model should be accompanied by the seed pass rate. This protocol does not remove seed fragility, but it makes the fragility explicit and prevents relation-only selection from silently choosing an unsafe residual direction.

4 Experiments

4.1 Setup

All runs use OpenCLIP ViT-B/16 with LAION2B weights. The image and text encoders are frozen; only the adaptation modules are trainable. The selected R028 configuration uses $K = 8$ residual channels, fixed residual gate $\alpha = 0.05$, $\lambda_{\text{anchor}} = 10$, $\lambda_{\Delta} = 0.01$, anchor ratio 0.75, and no targeted relational loss. Remote jobs ran on RTX A6000 GPUs with no wandb logging.

The staged relation evaluation combines ARO/SugarCrep-style relation and compositional counterfactual suites. Guardrail checks use COCO retrieval with 1,000 images and 5,002 captions, plus ImageNet zero-shot with 5,000 examples. External follow-ups use Winoground’s 400 examples and Flickr30k retrieval with 1,000 images. The pilot gate used during experimentation required relation accuracy above the plain adapter, COCO image-to-text R@1 at least 0.70, and ImageNet top-1 at least 0.55.

The evaluation is organized by run families rather than isolated best checkpoints. R003 is frozen OpenCLIP. R005 is the plain adapter baseline. R011 and R014 are representative hard-token bottlenecks that replace the embedding path. R019, R025, and R027 are full RRB seed checks. R023, R028, R031, R032, and R035–R039 remove the targeted loss under different anchoring settings. R024 replaces structured relation counterfactuals with generic hard negatives. R030 is the detailed error-analysis diagnostic export used for the spatial failure taxonomy. R033 and R040 remove the multi-channel bottleneck from the residual interface. R034, R041, R043, and R045 are follow-ups for external validity, mechanism, orthogonal projection, and Flickr30k retrieval.

The reported metrics are intentionally mixed. Relation accuracy measures the targeted failure mode, calculated as the simple average accuracy across all evaluated ARO and

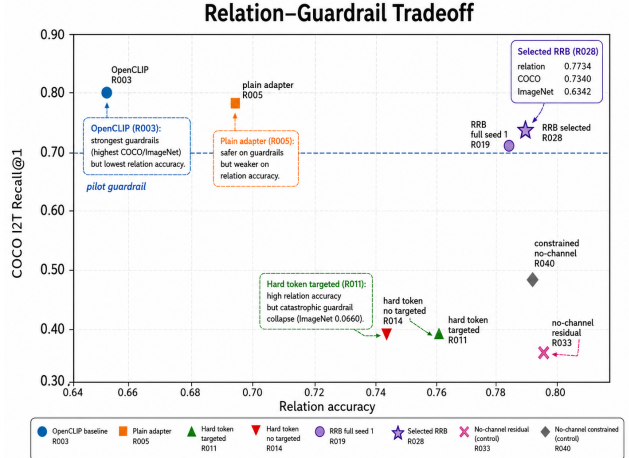


Figure 2: Relation accuracy versus COCO image-to-text retrieval for representative evaluated systems. The selected R028 checkpoint improves relation accuracy relative to the plain adapter while staying above the project COCO guardrail; hard-token and no-channel controls show that relation gains can also be obtained by damaging retrieval geometry.

SugarCrep sub-suites. COCO image-to-text R@1 measures whether the image embedding still retrieves the correct caption from a small held-out retrieval pool. ImageNet top-1 measures zero-shot category transfer. Residual magnitude and cosine-to- z_0 summarize how much the adapted vector moves, but they are diagnostic rather than decisive. Winoground text, image, and group scores test a different binding setting, and Flickr30k checks whether the retrieval tradeoff transfers beyond the COCO guardrail split.

All numbers in this section come from the consolidated local experiment report and the copied remote-output summaries. The draft does not rerun the GPU training jobs during paper generation. That matters for interpretation: the paper is a faithful analysis of the existing run record, not a new benchmark campaign with exhaustive hyperparameter search.

4.2 Main Tradeoff

Table 1 and Figure 2 summarize the main tradeoff. In Figure 2, the selected configuration is highlighted with a star marker to distinguish it from other seed variations and baseline controls. OpenCLIP has the strongest broad guardrails among the core baselines, but lower relation accuracy. The plain adapter improves relation from 0.6564 to 0.6956 with a small guardrail tax. Hard token bottlenecks move relation further but collapse both guardrails: the targeted hard-token bottleneck (R011) reaches relation 0.7488 with COCO R@1 0.3920 and ImageNet top-1 0.0660.

RRB demonstrates a distinct Pareto-style tradeoff compared with replacement bottlenecks. The first full RRB seed

Table 1: Main relation/anchor-preservation comparison. Relation accuracy is measured on staged ARO/SugarCrepe-style relation suites. COCO is image-to-text Recall@1 on the 1k-image guardrail split. Residual is the mean relative magnitude of the learned residual branch where available.

Run	System	Relation	Δ vs R005	COCO R@1	ImageNet	Residual	Notes
R003	OpenCLIP	0.6564	–	0.7990	0.6670	–	Frozen OpenCLIP baseline
R005	plain adapter	0.6956	0.0000	0.7790	0.6438	–	Plain adapter baseline
R011	hard token targeted	0.7488	0.0532	0.3920	0.0660	–	Catastrophic guardrail degradation
R014	hard token no targeted	0.7332	0.0376	0.4110	0.0602	–	Catastrophic guardrail degradation
R019	RRB full seed 1	0.7697	0.0741	0.7190	0.6152	0.1328	Clears pilot guardrails
R028	RRB no targeted, stronger anchor	0.7734	0.0778	0.7340	0.6342	0.0804	Selected configuration
R033	no-slot residual adapter control	0.7825	0.0869	0.3740	0.3308	0.2062	Catastrophic guardrail degradation
R040	constrained no-slot residual control	0.7773	0.0817	0.4820	0.4798	0.0969	Large retrieval/zero-shot degradation

check (R019) reaches relation 0.7697 while passing the pilot COCO and ImageNet thresholds. The selected no-targeted stronger-anchor checkpoint (R028) simplifies the method by removing the targeted loss and strengthening anchoring; it reaches relation 0.7734, COCO R@1 0.7340, and ImageNet top-1 0.6342. This is still an alignment tax relative to OpenCLIP and the plain adapter, but it is not the catastrophic forgetting observed in the replacement bottlenecks.

The no-channel controls sharpen the interpretation. The unconstrained no-slot residual adapter (R033) reaches relation 0.7825, higher than R028, but collapses COCO to 0.3740 and ImageNet to 0.3308. The constrained no-slot residual control (R040) uses the same no-channel idea with stronger regularization and reduces residual magnitude, but COCO remains only 0.4820 and ImageNet 0.4798. These controls show why relation accuracy cannot be optimized alone. They also support the architectural role of the multi-channel interface: a residual path without that bottleneck can find relation-improving directions that are incompatible with broad CLIP behavior.

The selected result is therefore a Pareto-style point, not a strict dominance claim. OpenCLIP and the plain adapter retain stronger retrieval. Some hard or unconstrained variants obtain relation scores close to or above R028. R028 is useful because it occupies a middle region that the earlier variants failed to reach: a substantial relation gain with retrieval and zero-shot scores still in a usable range. That is the paper’s main empirical claim.

4.3 What Training Signal Matters?

Table 2 separates structured counterfactual supervision from the targeted relational loss. The generic-hard-negative control (R024) replaces structured relational counterfactuals with generic hard negatives. It preserves COCO and ImageNet nearly perfectly, but relation accuracy drops

Table 2: Structured counterfactuals, not the targeted relational loss $L_{targeted}$, explain the retained relation gain.

Run	Variant	Struct. CF	$L_{targeted}$	Relation	COCO
R023	no-targeted, weak anchor	yes	0	0.7922	0.6820
R024	generic hard negatives	no	1	0.6736	0.7990
R028	selected stronger-anchor	yes	0	0.7734	0.7340
R029	matched targeted control	yes	1	0.7731	0.7280

to 0.6736, below the plain adapter baseline. In contrast, the no-targeted weak-anchor run (R023) and the selected no-targeted stronger-anchor run (R028) use structured counterfactuals with $L_{targeted} = 0$ and retain large relation gains. The matched stronger-anchor targeted control (R029) adds the targeted loss back under the same stronger-anchor setting and does not improve the selected tradeoff. The result is a useful anti-claim: the targeted loss sounded plausible but is not the supported mechanism.

This ablation is central because it prevents two overstatements. First, the gain is not explained by hard-negative training in general. If that were the explanation, R024 would have improved relation accuracy while preserving guardrails. It did the opposite: it preserved guardrails and lost the relation effect. Second, the gain is not evidence that targeted channel supervision discovers relation-specialized channels. R023 and R028 remove the targeted term entirely. R028 is the cleaner paper-facing method because it keeps the structured relation signal and the anchor terms while dropping a mechanism that the evidence does not need.

The comparison between the no-targeted weak-anchor run (R023) and the selected stronger-anchor run (R028) also shows why model selection cannot use relation accuracy alone. R023 reaches relation 0.7922, higher than R028, but COCO falls to 0.6820. R028 gives up some relation accuracy

Table 3: Seed summaries for RRB run families. Full RRB refers to R019/R025/R027, anchor-10 no-targeted refers to the R028/R031/R032 family, and anchor-20 no-targeted refers to R035–R039. Values are mean $\pm$ sample standard deviation.

Group	Relation	COCO R@1
full RRB	0.7768 ± 0.0062	0.6927 ± 0.0294
anchor-10 no-targeted	0.7782 ± 0.0042	0.7110 ± 0.0332
anchor-20 no-targeted	0.7789 ± 0.0032	0.7092 ± 0.0232

and passes the pilot guardrail. This tradeoff is exactly what the stronger anchor was meant to expose. The selected checkpoint is not the maximum relation model in the run history; it is the simplest relation-positive model that clears the explicit preservation gate.

4.4 Seed and Anchoring Stability

The seed summaries in Table 3 show stable relation gains and unstable retrieval retention. Full RRB over R019/R025/R027 has mean relation 0.7768 ± 0.0062 , but mean COCO R@1 0.6927 ± 0.0294 , with only one of three runs clearing the COCO threshold. No-targeted anchor-10 runs R028/R031/R032 have mean relation 0.7782 ± 0.0042 and mean COCO 0.7110 ± 0.0332 . Anchor-20 runs R035–R039 reduce residual drift in the consolidated diagnostics and keep relation stable, but their mean COCO remains 0.7092 ± 0.0232 . The method therefore supports relation gains across seeds, not robust retrieval preservation across seeds.

The anchor-20 group is especially informative. It reduces mean residual magnitude to 0.0666 and increases mean cosine-to- z_0 to approximately 0.9975, but this does not translate into a clearly higher COCO mean than the anchor-10 group. The best anchor-20 external-retrieval candidate (R038) has the highest COCO score in that group and is used for the Flickr30k follow-up, yet other anchor-20 seeds still miss or nearly miss the COCO threshold. The result argues against a simple residual-size selection rule. Stronger anchoring can reduce drift, but relation-preserving directions and retrieval-preserving directions are not identical.

For a deployment-oriented paper, this would be a serious blocker. For this paper’s more limited claim, it is part of the result. RRB gives a repeatable relation gain under multiple seeds and anchor settings, but retrieval preservation remains stochastic and incomplete. The right summary statistic is therefore not only the selected R028 point; it is the seed-group pass rate and variance around that point.

This is why the gate in Section 3.6 is a validation protocol rather than a guarantee. A relation-positive checkpoint is not considered usable until it also passes the COCO and ImageNet guardrails, and a run family is not considered stable unless its pass rate is reported. Under that stricter reading, R028 is evidence that a useful residual direction exists, while

the seed table shows that the current training recipe does not find such directions reliably enough for deployment.

4.5 External Validity and Mechanism Checks

Table 4 gives the boundary checks. The Winoground external-validity run (R034) does not support broad binding generalization: R028 scores 0.2400 text, 0.1100 image, and 0.0700 group, while OpenCLIP scores 0.2775, 0.1075, and 0.0825. The Flickr30k retrieval follow-up (R045) confirms an external image-to-text retrieval tax, meaning a drop in image-to-text R@1 relative to OpenCLIP: R028 drops from 0.8620 to 0.8260, while R038 improves that to 0.8440 but still does not match OpenCLIP. The channel-responsibility probe (R041) is diffuse, with entropy near 1.0000 and top-1 concentration near 0.126, so the paper does not claim semantic channel specialization. The orthogonal-projection preservation ablation (R043) preserves COCO and ImageNet better but fails the relation target, so it remains a cited ablation or future direction rather than a new headline.

The Winoground result is the cleanest warning against overgeneralization. R028 slightly improves image score by 0.0025 over OpenCLIP but loses text and group score. A broad binding claim would require improvement on the group score, not only on staged ARO/SugarCreme-style edits. The paper therefore treats Winoground as negative external-validity evidence. This is not surprising: the training signal emphasizes edited captions and margin ranking, while Winoground requires matching two captions and two images in a hand-curated setting where both visual and textual grounding must be correct.

Flickr30k gives a different warning. It is not a relation benchmark; it is an external retrieval distribution. The R028 image-to-text drop shows that the COCO guardrail is not a complete proxy for retrieval retention. R038 narrows the drop but does not close it. The text-to-image numbers are much closer to OpenCLIP, so an asymmetric retrieval tax is a plausible future-work hypothesis, not a claim established by the current small follow-up.

The responsibility probes close off the interpretability story. A useful semantic-channel result would show concentrated responsibility, consistent edited-token overlap, or stable channel assignments across examples. The observed entropy and top-1 concentration are close to diffuse. This does not invalidate the bottleneck as an architectural prior, but it does invalidate a stronger slot-semantics explanation. The final method name and captions therefore avoid promising interpretable slots.

5 Analysis and Limitations

5.1 Residual Size Is Not Sufficient

Residual magnitude is useful diagnostic information, but it is not a standalone safety predictor. The no-targeted weak-anchor run (R023) has low residual magnitude and high cosine to the frozen embedding, yet misses the COCO gate.

Table 4: External-validity and mechanism checks bound the claim. These follow-ups strengthen the limitation story rather than changing the main positive result.

Check	Result	Interpretation
<i>Downstream and external retrieval checks</i>		
Winoground R034	R028 text/image/group 0.2400/0.1100/0.0700 vs Open-CLIP 0.2775/0.1075/0.0825	Text and group scores drop while image score is nearly flat; no broad binding claim.
Flickr30k R045	R028 I2T R@1 0.8260 vs Open-CLIP 0.8620; R038 0.8440	External retrieval tax persists, smaller for anchor-20 R038.
<i>Mechanistic and preservation probes</i>		
Slot responsibility R041	entropy 1.0000, top-1 concentration 0.1259, edited overlap 0.1397	Diffuse responsibility; no interpretable semantic-channel claim.
ROP R043	ablation relation 0.7441, COCO ImageNet 0.6566	Better preservation but relation gate fails; ROP remains an ablation/future direction.

The selected stronger-anchor run (R028) improves COCO under a similar small residual budget, while the matched targeted control (R029) has a larger residual and slightly lower COCO than R028. Anchor-20 runs reduce residual drift further, but do not solve COCO seed fragility. The direction and objective of the residual matter, not only its norm.

This point matters because residual methods are often described as safe by construction. RRB is safer than the hard replacement bottlenecks in this run history, but the residual form alone is not the safety argument. A residual can still move examples in directions that damage nearest-neighbor structure; conversely, a moderately larger residual can be acceptable if it aligns with relation-sensitive distinctions while preserving local neighborhoods. Residual scale, anchor mimicry, and the multi-channel interface jointly improve the tradeoff, but none guarantees preservation.

The selected checkpoint is best understood as a compromise between three pressures: separating edited captions, keeping clean examples near their frozen embeddings, and keeping the correction small. When these pressures are balanced well, as in R028, relation accuracy improves with bounded degradation. When one pressure dominates or the interface is too unconstrained, the model either fails to learn relation sensitivity or damages retrieval.

5.2 Generalization Gap

The main positive result lives on staged relation/counterfactual suites. Those suites are useful because they isolate caption edits, but they are not equivalent to natural compositional reasoning. Winoground is hand-curated and requires paired text-image discrimination; the selected R028 checkpoint does not improve it. Flickr30k uses a different retrieval distribution and shows an external image-to-text retrieval tax. A plausible explanation is that structured counterfactual training teaches sensitivity to benchmark-style relational edits without fully learning the broader semantics needed for natural binding. This is a limitation, not a hidden success.

The per-source relation results reinforce this interpretation. R028 is strong on several SugarCrepe replacement categories and word-order checks, but ARO visual relation remains difficult, with left/right predicates dominating the failure taxonomy. This pattern suggests that the model is not simply acquiring a general spatial calculus. It learns useful distinctions for many edited captions, but some predicates remain close to the frozen model’s blind spots. The improvement is therefore best described as counterfactual relation sensitivity, not full spatial reasoning.

The gap also affects how future data should be designed. Adding more edited captions of the same type may continue to improve the staged benchmark without improving Winoground or external retrieval. A stronger next experiment would include validation splits that are not generated by the same edit templates, paired-image binding checks, and retrieval datasets beyond COCO. The current paper uses external checks after the main run history to expose this issue; it does not claim to have solved it.

The objective itself explains part of the gap. Eq. 2 mostly contrasts one image with a positive caption and an edited caption. Winoground instead requires a two-image, two-caption decision where both cross-pair assignments must be correct. A natural next training signal is therefore image-pair contrast: construct paired images and paired captions that mimic the Winoground structure, then require the residual to improve the full assignment rather than only a single-image caption margin.

5.3 Failure Taxonomy

The error-analysis diagnostic export (R030) for the selected R028 checkpoint shows that left/right spatial predicates remain difficult, and SugarCrepe object and attribute swaps still account for many errors. These failures are consistent with the external checks: RRB changes the relation-vs-preservation tradeoff, but it is not a complete spatial reasoning solution.

The margin information in the diagnostic report is also useful. The left/right failures have margins near zero, which

means many examples are not confidently wrong by a large amount; they are ambiguous under the adapted similarity function. This is different from a failure where the model strongly prefers the counterfactual caption. Small margins suggest that additional supervision, better image grounding, or a more targeted residual direction constraint might move some examples. They do not by themselves show that the current model has the right abstraction.

Object and attribute swaps are a separate failure class. These are not purely spatial, and they indicate that the relation objective interacts with broader compositional binding. A method that only improves left/right predicates would not be enough. Conversely, a method that improves object and attribute swaps while losing retrieval may simply be changing the embedding geometry too broadly. The failure taxonomy therefore acts as a check on whether the residual is correcting the intended errors or merely shifting benchmark scores.

5.4 Missing Baselines

The available ablation suite compares RRB to OpenCLIP, a plain adapter, hard bottlenecks, generic hard negatives, no-targeted variants, no-channel residual controls, stronger anchoring, ROP, Winoground, and Flickr30k. It does not include standard PEFT baselines such as LoRA, text-only tuning, or vision-only tuning. While RRB shows stronger retention than naive residual controls in this study, its performance relative to widely used PEFT methods remains to be characterized in future work [4].

This missing-baseline caveat should change how the paper is read. The result is enough to support an internal method analysis: among the implemented designs, bounded multi-channel residual adaptation gives the best relation-preservation compromise. It is not enough to claim state-of-the-art adaptation. A direct PEFT comparison would need matched training data, the same guardrail metrics, seed repeats, and external retrieval checks. Without that, the paper should avoid language such as “outperforms PEFT” or “dominates adapter baselines.”

The ROP follow-up is similarly bounded. Orthogonal or principal-subspace constraints are plausible preservation mechanisms, but they are close to existing preservation-oriented adaptation ideas rather than a separate novelty claim. The orthogonal-projection ablation (R043) preserves COCO and ImageNet better than some RRB variants but fails the relation target. That makes it useful evidence about residual direction, not a replacement headline. The lack of relation gain suggests that naive orthogonal constraints may be too restrictive for this training signal, leaving principal-subspace projection as a future guardrail.

5.5 Challenges and Resolutions

The first challenge was that relation accuracy was easy to improve by damaging the embedding geometry. The hard-token bottleneck runs and the no-channel residual controls expose this failure mode: they raise relation scores, but their

COCO and ImageNet guardrails collapse or remain far below the selected RRB checkpoint. The resolution was to treat relation accuracy as only one side of model selection and require explicit COCO and ImageNet guardrails before a checkpoint could become paper-facing evidence.

The second challenge was that small residuals were not automatically safe. The no-targeted weak-anchor run (R023) keeps residual magnitude low but still misses the COCO gate, while the selected stronger-anchor run (R028) finds a better relation-preservation compromise. The project therefore added stronger feature anchoring and reported seed-group pass rates rather than claiming that residual magnitude alone controls safety.

The third challenge was external validity. Winoground and Flickr30k weakened the broadest possible interpretation of the result: RRB improves staged counterfactual relation sensitivity, but it does not show broad binding transfer or lossless retrieval preservation. The resolution is the paper’s conservative claim boundary: present RRB as a guarded residual-interface tradeoff analysis, not as a solved compositional-reasoning method.

5.6 Threats to Validity

Several threats remain. First, the guardrail datasets are sampled: COCO uses 1,000 retrieval images and ImageNet uses 5,000 examples. These checks catch catastrophic collapse, but they are not a full evaluation suite. Second, the relation suites are staged counterfactual benchmarks and may share template structure with the training signal. Third, the run history is a project campaign rather than a fully randomized hyperparameter study. Some comparisons are tightly matched, such as R028 versus R029, while others compare design families discovered over time.

Fourth, the paper evaluates embedding behavior, not downstream user tasks. Better relation sensitivity in a dual encoder may help retrieval, filtering, or ranking, but this draft does not test an end application. Fifth, the method is tied to OpenCLIP ViT-B/16 with LAION2B weights. Other backbones may have different baseline relation sensitivity and different tolerance for residual adaptation. Sixth, the external validity checks are negative or mixed. That makes the claim honest, but it also means that the method should not be presented as a broad compositional reasoning solution.

These threats do not erase the main result. They define its scope. The evidence supports a bounded statement about adding relation sensitivity through a residual interface while avoiding the worst guardrail failures in this run family. It does not support stronger claims about universal retrieval preservation, semantic channel discovery, or broad binding generalization.

5.7 Practical Implications

The practical lesson is to evaluate relation adaptation with guardrails from the beginning. If the project had selected only by relation accuracy, the hard bottlenecks

and no-channel controls would have looked attractive. Their retrieval and zero-shot metrics show that they are poor candidates for a CLIP-compatible method. Conversely, if the project had selected only by COCO and ImageNet preservation, the generic-hard-negative run would have looked safe while failing to address the relation problem. The useful region is visible only when both sides are measured together.

For future experiments, the highest-value additions are not more small variations of the selected checkpoint. They are matched PEFT baselines, larger external retrieval checks, and validation sets whose relation edits differ from the training templates. A better residual-direction constraint is also promising, but it should be evaluated as a guardrail mechanism rather than introduced as a novelty claim before the evidence exists.

5.8 Claim Boundary

The supported claim is that anchored residual adaptation improves relation sensitivity while bounding, but not eliminating, retrieval and zero-shot degradation. The unsupported claims are just as important: RRB does not prove broad Winoground binding generalization, does not preserve retrieval losslessly, does not show semantic slot specialization, and does not establish orthogonal projection as a novel method.

6 Conclusion

RRB keeps the frozen OpenCLIP embedding path and learns a bounded relational residual from parser-free token channels. In the available runs, this design improves staged relation accuracy with far less guardrail damage than hard replacement bottlenecks and no-channel residual controls. The key supervision is structured relational counterfactual training under residual anchoring; the targeted relational loss is not necessary for the supported effect.

The result should be read as a tradeoff analysis. COCO retention remains seed-fragile, and that fragility is itself part of the finding rather than a minor implementation detail. Winoground is not improved, Flickr30k shows an external retrieval tax, and channel responsibility is diffuse. Future work should compare against standard PEFT baselines, test broader retrieval and binding distributions, and develop better constraints on residual direction rather than residual magnitude alone.

The main value of the study is the failure-aware path it establishes. It shows that relation signal can be added to a frozen CLIP-style encoder without replacing the embedding path, but it also shows that preservation must be measured directly and repeatedly. The strongest next version of this work would keep the conservative claim boundary, add matched PEFT and larger retrieval baselines, and test whether residual-direction constraints can make the relation gain less seed-fragile.

7 Reproducibility

The project repository is available at https://github.com/sunTr1aL/CS510_Project. The repository contains the experiment scaffold, run registry, launch scripts, compact result summaries, generated figure/table assets, paper source, and tests. The full raw benchmarks, checkpoints, and remote training outputs are not committed because they are large generated artifacts or externally governed datasets; the checked-in summaries are the source for the reported paper tables and figures.

The main experiments use frozen OpenCLIP ViT-B/16 with LAION2B weights. The selected R028 configuration uses $K = 8$ residual channels, fixed residual gate $\alpha = 0.05$, $\lambda_{\text{anchor}} = 10$, $\lambda_{\Delta} = 0.01$, anchor ratio 0.75, and no targeted relational loss. The remote training scripts record RTX A6000 GPU runs in the `relclip` environment with `wandb` disabled. Defaults in `scripts/run_rrb_batch.sh` use 2,048 training examples, 512 evaluation examples per relation key, 5 epochs, batch size 64, learning rate 10^{-4} , COCO retrieval over 1,000 images, and ImageNet zero-shot over 5,000 examples.

Local verification can be run with `python -m unittest tests.test_experiment_scaffold`. Paper assets can be regenerated with `python figures/generate_paper_assets.py`. The LaTeX source in `paper/` builds with `pdflatex`, `bibtex`, and two final `pdflatex` passes. Remote scripts include `scripts/launch_active_rrb_remote.sh`, `scripts/run_rrb_batch.sh`, `scripts/run_winoground_r034.sh`, and `scripts/run_topvenue_batch.sh`.

8 Team Member Contributions

Zixuan Sun (zixuans8, zixuans8@illinois.edu) served as team coordinator and contributed to the RRB method design, experiment implementation, remote run orchestration, result consolidation, and final report preparation.

Dehao Wu (dehaowu2, dehaowu2@illinois.edu) contributed to benchmark and evaluation work, including dataset staging, retrieval/zero-shot guardrail checks, and interpretation of the COCO, ImageNet, Winoground, and Flickr30k results.

Tianyi Wang (tianyiw5, tianyiw5@illinois.edu) contributed to ablation analysis, result summarization, and the comparison of structured counterfactual training against generic hard negatives, targeted-loss variants, and no-channel controls.

Yifei Wang (yifeiw16, yifeiw16@illinois.edu) contributed to related-work synthesis, limitation analysis, figure/table review, and reproducibility packaging for the final ACM-style report.

References

- [1] Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Danny Driess, Pete Florence, Dorsa Sadigh, Leonidas J. Guibas, and Fei Xia. 2024. SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. doi:10.1109/CVPR52733.2024.01370

- [2] Biao Gong, Shuai Tan, Yutong Feng, Xiaoying Xie, Yuyuan Li, Chaochao Chen, Kecheng Zheng, Yujun Shen, and Deli Zhao. 2024. UKnow: A Unified Knowledge Protocol with Multimodal Knowledge Graph Datasets for Reasoning and Vision-Language Pre-Training. doi:10.48550/arXiv.2302.06891
- [3] Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Aniruddha Kembhavi, and Ranjay Krishna. 2023. SugarCrepe: Fixing Hackable Benchmarks for Vision-Language Compositionality. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*. <https://openreview.net/forum?id=Jsc7WSCZd4>
- [4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.
- [5] Gabriel Ilharco, Mitchell Wortsman, Ross Wightman, Cade Gordon, Nicholas Carlini, Rohan Taori, Achal Dave, Vaishaal Shankar, Hongseok Namkoong, John Miller, Hannaneh Hajishirzi, Ali Farhadi, and Ludwig Schmidt. 2021. *OpenCLIP*. doi:10.5281/zenodo.5143773
- [6] Aneesh Komanduri, Karuna Bhaila, and Xintao Wu. 2025. CausalVLBench: Benchmarking Visual Causal Reasoning in Large Vision-Language Models. doi:10.48550/arXiv.2506.11034
- [7] Hao Liu, Yanni Ma, Yan Liu, Haihong Xiao, and Ying He. 2024. 3D Scene Graph Guided Vision-Language Pre-training. doi:10.48550/arXiv.2411.18666
- [8] Xin Liu, Yi He, Yiu-Ming Cheung, Xing Xu, and Nannan Wang. 2024. Learning Relationship-Enhanced Semantic Graph for Fine-Grained Image-Text Matching. *IEEE Transactions on Cybernetics* 54, 2 (2024). doi:10.1109/TCYB.2022.3179020
- [9] Yuankun Liu, Xiang Yuan, Haochen Li, Zhijie Tan, Jinsong Huang, Jingjie Xiao, Weiping Li, and Tong Mo. 2024. SEM-Scene: Semantic-Consistency Enhanced Multi-Level Scene Graph Matching for Image-Text Retrieval. *ACM Transactions on Multimedia Computing, Communications, and Applications* 20, 8, Article 261 (2024). doi:10.1145/3664816
- [10] Yue Lu, Shizhou Zhang, De Cheng, Yinghui Xing, Nannan Wang, Peng Wang, and Yanning Zhang. 2024. Visual Prompt Tuning in Null Space for Continual Learning. doi:10.48550/arXiv.2406.05658
- [11] Mao-Lin Luo, Zi-Hao Zhou, Yi-Lin Zhang, Yuanyu Wan, Min-Ling Zhang, and Tong Wei. 2026. KeepLoRA: Continual Learning with Residual Gradient Adaptation. In *International Conference on Learning Representations*.
- [12] Wentao Ma, Xiaoqian Wu, Shaofan Chen, Wei Liu, Shan Zhao, Qian Wan, and Lichuan Gu. 2026. TSGR2: Image-Text Matching via Triple-Level Scene Graph Relation Reasoning. *Applied Soft Computing* 187 (2026), 114323. doi:10.1016/j.asoc.2025.114323
- [13] Chancharik Mitra, Brandon Huang, Trevor Darrell, and Roei Herzig. 2024. Compositional Chain-of-Thought Prompting for Large Multimodal Models. doi:10.48550/arXiv.2311.17076
- [14] Tiantian Peng, Yuyang Liu, Shuo Yang, Qiuhe Hong, and Yonghong Tian. 2025. GNSP: Gradient Null Space Projection for Preserving Cross-Modal Alignment in VLMs Continual Learning. doi:10.48550/arXiv.2507.19839
- [15] Jingyang Qiao, Zhizhong Zhang, Xin Tan, Yanyun Qu, Wensheng Zhang, Zhi Han, and Yuan Xie. 2024. Gradient Projection For Continual Parameter-Efficient Tuning. doi:10.48550/arXiv.2405.13383
- [16] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *International Conference on Machine Learning*.
- [17] Zanxi Ruan, Songqun Gao, Qiuyu Kong, Yiming Wang, and Marco Cristani. 2026. StructXLIP: Enhancing Vision-Language Models with Multimodal Structural Cues. doi:10.48550/arXiv.2602.20089
- [18] Cheng Shi, Yizhou Yu, and Sibe Yang. 2026. Vision Transformers Need More Than Registers. doi:10.48550/arXiv.2602.22394
- [19] Lingyun Song, Wenqing Du, Xiaolin Han, Xinbiao Gan, Xiaoqi Wang, and Xuequn Shang. 2025. Cross-modal Multi-Relational Graph Reasoning: A Novel Model for Multimodal Textbook Comprehension. *Information Fusion* 120 (2025), 103082. doi:10.1016/j.inffus.2025.103082
- [20] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. 2022. Winoground: Probing Vision and Language Models for Visio-Linguistic Compositionality. doi:10.48550/arXiv.2204.03162
- [21] Unknown Authors. 2026. RACA-CLIP: Relation-Aware Compositional Alignment for CLIP. Local literature copy; bibliographic metadata incomplete in project wiki.
- [22] Sijin Wang, Ruiping Wang, Ziwei Yao, Shiguang Shan, and Xilin Chen. 2019. Cross-modal Scene Graph Matching for Relationship-aware Image-Text Retrieval. doi:10.48550/arXiv.1910.05134
- [23] Zhaotian Weng, Haoxuan Li, Xin Eric Wang, Kuan-Hao Huang, and Jieyu Zhao. 2026. What's Missing in Vision-Language Models? Probing Their Struggles with Causal Order Reasoning. doi:10.48550/arXiv.2506.00869
- [24] Fei Wu, Jia Hu, Geyong Min, and Shiqiang Wang. 2026. Efficient Orthogonal Fine-Tuning with Principal Subspace Adaptation. In *International Conference on Learning Representations*.
- [25] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. 2022. When and why vision-language models behave like bags-of-words, and what to do about it? doi:10.48550/arXiv.2210.01936
- [26] Yongshuo Zong, Bingchen Zhao, Letian Zhang, and Timothy Hospedales. 2024. Benchmarking Multi-Image Understanding in Vision and Language Models: Perception, Knowledge, Reasoning, and Multi-Hop Reasoning. doi:10.48550/arXiv.2406.12742